

A configurable global context reconstruction hybrid detector for enhanced small object detection in UAV aerial imagery

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ABSTRACT

To address the challenge of balancing detection accuracy and efficiency for small objects in complex aerial scenes, we propose a Configurable Global Context Reconstruction Hybrid Detector (GCRH) to enhance overall detection performance. The GCRH framework consists of three key components. First, the Efficient Re-parameterized Encoder (ERE) reduces the computational overhead of multi-head self-attention through re-parameterization while maintaining the integrity and independence of global-local feature interactions. Second, the Global-Aware Feature Pyramid Network (GAFPN) reconstructs and injects global contextual semantics, cascading selective feature fusion to distribute this semantic information across feature layers, thereby alleviating small-object feature degradation and cross-level semantic inconsistency. Finally, two configurable model variants are provided, allowing the control of high-resolution feature layers to balance detection accuracy and inference efficiency. Experiments on the VisDrone2019 and TinyPerson datasets demonstrate that GCRH achieves an effective trade-off between precision and efficiency, validating its applicability to small object detection in aerial imagery. The code is available at: <https://github.com/Mundane-X/GCRH>.

1. Introduction

With the rapid advancement of unmanned aerial vehicle (UAV) technology and the continuous reduction in manufacturing costs, UAVs have been increasingly deployed in a wide range of fields, including environmental monitoring, disaster response, and urban planning (Dimtopoulos et al., 2014). These applications highlight the enormous potential of UAVs in real-world scenarios. In most UAV-based tasks, real-time object detection serves as a fundamental prerequisite for autonomous perception and immediate decision-making. In contrast, offline detection is often limited by high latency, reliance on communication links, and delayed response strategies, making it unsuitable for time-sensitive applications (Sun et al., 2022). However, due to UAVs' unique flight altitudes and oblique imaging perspectives, aerial imagery frequently contains numerous small objects embedded in complex backgrounds, posing significant challenges for conventional object detection algorithms (Leng et al., 2022). Achieving high-accuracy real-time detection under such conditions remains a core technical bottleneck in UAV visual perception. Addressing this issue is critical for advancing UAV

intelligence and enabling their widespread deployment in real-world scenarios.

Modern object detectors have achieved remarkable success in generic detection tasks; however, their performance often degrades significantly when dealing with small-scale objects. This degradation is primarily due to the limited number of pixels representing small objects in an image, which makes them highly susceptible to the degradation of discriminative features during downsampling and the inconsistency of semantic information across feature levels. As a result, detectors often fail to capture the fine-grained and distinctive cues necessary for small object recognition in complex scenes. One of the main causes of this performance gap lies in the fact that traditional feature extraction and fusion methods are not explicitly designed for enhancing small object representations. To address this issue, recent approaches have attempted to improve performance by employing deeper backbone networks, additional network layers, or more sophisticated attention mechanisms. However, such methods often introduce considerable computational overhead, leading to a reduction in inference speed.

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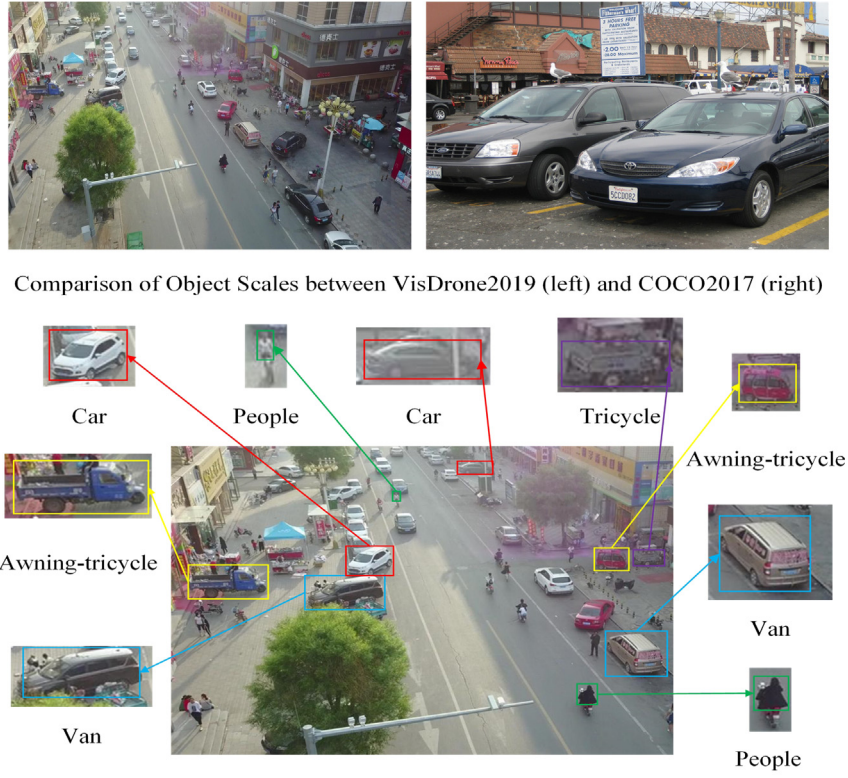


Fig. 1. The difficult object samples in VisDrone2019-DET.

Furthermore, multi-scale architectures like the Feature Pyramid Network (Lin et al., 2017) have been widely adopted to combine low-level details with high-level semantics. Yet, naive fusion strategies frequently result in feature loss and semantic misalignment across different scales. This not only compromises essential feature information but also hampers the effective alignment of representations across pyramid levels. In summary, existing methods often suffer from global contextual information loss, cross-level semantic inconsistency, and high computational complexity, which severely limit their effectiveness in UAV-based aerial object detection.

In UAV-based aerial imagery, objects exhibit substantial variation in both scale and visual characteristics. It is common to observe large structures, small vehicles, and pedestrians coexisting within a single frame. Due to the flight altitude and randomly rotated top-down perspectives of UAVs, small objects often occupy less than 1% of the image area and experience significant distortion in visual shape. As illustrated in Fig. 1, the VisDrone2019 dataset (Zhu et al., 2021) contains a large number of small objects (smaller than 30 px) and considerable intra-class scale variation, even among objects of the same category. These properties further confirm that the unique imaging conditions of UAVs adversely affect object scale and spatial distribution, thereby increasing the difficulty of detection. Moreover, aerial imagery is susceptible to various external interference factors, such as altitude, lighting, and occlusion, which result in high inter-class similarity (e.g., between Car and Van) and substantial intra-class variability (e.g., Awning-tricycle). Consequently, capturing accurate semantic information becomes critical for enhancing object discriminability in UAV-based detection tasks.

To address the aforementioned challenges, we propose a Global Context Reconstruction Hybrid detector (GCRH), which integrates the design principles of RT-DETR (Zhao et al., 2024a) and feature pyramid architectures within a unified CNN-Transformer framework. The

proposed GCRH aims to enhance the detection performance of small objects, improve real-time inference speed, and maintain a balanced trade-off between accuracy and efficiency. The main contributions of this paper are summarized as follows:

1. An Efficient Reparameterization Encoder (ERE) is proposed to replace the computationally intensive conventional Transformer encoder. The ERE employs linear projection layers to encode queries and keys, while cross-layer connections with adaptive scaling factors are introduced to dynamically balance global and local token dependencies. This design enables logical and efficient token interactions with substantially reduced computational overhead, thereby improving small-object feature representation while maintaining high inference efficiency. Unlike Transformer structures that mainly emphasize query mechanisms or training strategies, the ERE achieves a fundamental architectural simplification of the encoder and enhances the stability of small-object feature expression.

2. A global-aware feature pyramid network is proposed to address the issues of critical feature loss and semantic inconsistency across feature levels. The context reconstruction module effectively propagates high-level abstract semantic information to lower-level feature maps, while the cascaded feature transmission path enhances the retention of essential features across different layers. The organic integration of these components facilitates improved global semantic consistency and local feature preservation across hierarchical levels, thereby enabling more robust multi-scale small object detection.

3. A configurable model architecture that allows the selective incorporation of high-resolution features from stage S_2 of the backbone network based on specific task requirements, enabling a flexible trade-off between detection accuracy and inference speed.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the proposed GCRH framework in detail. Section 4 reports experimental results and comparative

analysis. Section 5 concludes the paper and discusses potential future research directions.

2. Related work

2.1. Detection frameworks for small objects in UAV imagery

In the domain of small object detection for UAV aerial imagery, many researchers have been inspired by region search strategies and adopted a “divide-and-conquer” paradigm to design general frameworks that transition from global to local processing. For instance, PeleeNet (Ozge Unel et al., 2019) partitions images into multiple subregions using a tiling approach, enabling the model to better learn spatial representations and enhance small object detection performance. However, this method may compromise the integrity of certain object features during subregion division. To address the challenges specific to UAV imagery, Liang et al. (2019) proposed FS-SSD, a dual-branch feature pyramid detector designed for small object detection. FS-SSD leverages deconvolution and average pooling operations to construct parallel feature hierarchies, and further refines the feature fusion pathway to better preserve small object characteristics, thereby improving detection accuracy. Yang et al. (2019) proposed ClusDet, which first employs CPNet and ScaleNet to locate object clusters with high spatial density and subsequently applies DetecNet for secondary detection within these regions. To reduce model complexity, Li et al. (2020) introduced DMNet, which utilizes pixel intensity to guide category proportion learning. This composite architecture integrates a density map generation module, an image cropping module, and a detector, highlighting spatial relationships among objects and achieving strong performance on aerial imagery datasets. Building upon this paradigm, GLSAN (Deng et al., 2020) and CRENet (Wang et al., 2020) refined the cluster prediction process, allowing more accurate cluster regions to be passed to fine detectors, thus improving detection performance. Huang et al. (2022) introduced UFPMP-Det, which consists of two key components: the Unified Foreground Packing module merges subregions from a coarse detector to reduce background noise, while the Multi-Proxy Detection Network models targets at a fine-grained level, achieving a balance between speed and accuracy. Yuan et al. (2023) proposed CFNet, a two-stage coarse-to-fine network based on imitation learning for small object detection. In the first stage, the Coarse-to-Fine Region Proposal Network (CRPN) employs dynamic anchor selection and cascaded regression to generate a large number of high-quality proposals for small objects. The second stage introduces a feature imitation branch to refine these proposals through a teacher–student mechanism, where the small-object features in the detection head (student) are guided to mimic those of larger objects or a more powerful teacher model. Benefiting from this design, CFNet achieved state-of-the-art performance on small-object benchmarks such as SODA-D and SODA-A, significantly surpassing baseline detectors. These studies highlight the importance of scale-sensitive architectures and multi-stage refinement strategies, which inspired the scale-aware design of our proposed detection framework.

Despite their effectiveness in small object detection, these approaches often perform suboptimally on densely populated scenes. To address this, ReDet (Han et al., 2021) incorporates rotation-equivariant networks and employs rotation-invariant RoI Align to estimate object orientation, leading to model compactness and improved detection accuracy. Xi et al. (2022) proposed FiFoNet, which adaptively selects multiscale features and background interference combinations, restores multiscale differentiability, and strengthens low-quality representations of small objects, yielding significant gains on UAV datasets. Zhang et al. (2023b) developed SGMFNet, which features a global–local guided module to effectively fuse contextual and fine-grained information while mitigating background clutter. A reverse residual enhancement module is also introduced before the detection head to improve feature extraction for small objects, resulting in strong performance on the

VisDrone2019 dataset. Du et al. (2023) presented CEASC, a detector optimized for resource-constrained UAV platforms. It integrates sparse convolution into the detection head and applies an adaptive multi-layer masking strategy to generate optimal foreground masks across feature maps of varying scales, enhancing both accuracy and efficiency. Finally, Zhu et al. (2024) proposed a hybrid approach that combines global multi-level perception with dynamic region aggregation. Leveraging self-attention for global context and a sparse feature selection strategy for local aggregation, their method effectively reduces noise and enhances fine-grained representations. Zhang et al. (2025a) proposed a method that integrates adaptive downsampling with a scale-enhanced detection head to improve tiny object detection in remote sensing imagery. Instead of employing a fixed image pyramid, the method adaptively downsamples feature maps—regions likely to contain tiny objects are processed at higher resolutions, while simpler areas are downsampled more aggressively to reduce computational cost. Moreover, a scale-enhanced detection head is introduced to strengthen feature responses at object-relevant scales, thereby improving the detector’s sensitivity to tiny instances. Incorporating semantic context has also been shown to greatly benefit tiny object detection. Yuan et al. (2025) introduced a SemDiff framework to assist oriented small object detection in aerial imagery. By explicitly injecting semantic cues and addressing the optimization challenges of small objects, SemDiff significantly improves detection performance on challenging benchmarks such as SODA-A and Tiny-DOTA, outperforming existing one-stage detectors and achieving accuracy comparable to two-stage models while maintaining higher efficiency. Another noteworthy contribution is the Scene Contextualized Detection Network (SCDNet) developed by Zhao et al. (2024b) for dense tiny object detection. SCDNet leverages global scene context to handle images containing hundreds of tiny instances. It incorporates a dedicated scene classification sub-network that predicts scene types and provides high-level contextual priors, which are then fed into a lightweight fusion module to adaptively modulate the detector’s feature representations—effectively informing the model of expected object distributions and background semantics.

Inspired by the aforementioned methods, this work also integrates multi-level features and attention mechanisms. However, it places greater emphasis on achieving a balance between computational efficiency and detection performance through architectural design. Specifically, we propose an Efficient Reparameter Encoder (ERE) based on a linear additive attention mechanism, which enhances feature stability while reducing computational complexity. In addition, a Global-Aware Feature Pyramid Network (GAFPN) is introduced to enable cross-level semantic fusion, thereby mitigating feature loss and reinforcing semantic consistency for small object representation. The overall architecture is designed to significantly lower computational cost while maintaining high accuracy and achieving strong real-time detection capabilities.

2.2. Hierarchical feature pyramids for efficient multi-scale detection

Feature pyramids are a technique for handling multi-scale features and improving object detection accuracy, with the core objective of enabling models to perform well across objects of various sizes by processing features from different levels. Building upon the traditional image pyramid method, the Feature Pyramid Network (FPN) (Lin et al., 2017) introduces a top-down feature propagation mechanism with lateral connections, combining high-level semantic features with low-level spatial features. This approach effectively reduces information loss found in traditional methods, enhancing the detection ability for small objects while maintaining accuracy for large objects. The Path Aggregation Network (PANet) (Liu et al., 2018) builds upon FPN by incorporating a path aggregation mechanism, which adds bottom-up feature propagation, strengthening the interaction between low-level and high-level feature maps. However, this improvement introduces higher computational costs, requiring longer training and inference times,

making real-time inference on resource-constrained hardware challenging. AugFPN (Guo et al., 2020) narrows the semantic gap between different scales of features through consistency supervision, residual feature enhancement, and soft RoI selection, minimizing information loss from high-level feature maps. BiFPN (Tan et al., 2020), proposed by Google, further optimizes multi-scale feature fusion efficiency with an efficient bidirectional feature fusion structure, effectively integrating cross-scale features. Additionally, BiFPN uses a weighted feature fusion mechanism to allocate weights to different layers, optimizing information flow. GraphFPN (Zhao et al., 2021) can adaptively match the intrinsic structure of an image and employs two types of local channel attention, effectively fusing cross-scale features. To improve computational efficiency, EfficientRepGFPN (Xu et al., 2022) introduces a repetitive grouping feature aggregation mechanism, enhancing feature fusion through multiple aggregation operations while optimizing computations via “grouping” to reduce redundancy. RepPAN (Li et al., 2022a) introduces a repeated path aggregation mechanism, ensuring more comprehensive information exchange between high and low-level features, maximizing the complementarity of features. Its adaptive feature fusion mechanism dynamically allocates weights based on the importance of features at different scales, achieving more accurate feature integration. AFPN (Yang et al., 2023) dynamically adjusts and optimizes the feature fusion process, overcoming the limitations of traditional FPN in multi-scale feature handling. AFPN dynamically selects the most suitable feature fusion method based on object scale and image features, relying more on low-level features for small objects and high-level features for large objects. PMVOS (Zhang et al., 2025f) employs a pyramidal Transformer architecture to capture multi-scale feature representations across consecutive frames, combined with an adaptive fusion strategy that integrates coarse- and fine-scale information. This design effectively balances accuracy and efficiency, enabling temporally consistent object segmentation. Notably, the pyramid Transformer with adaptive fusion preserves both global contextual cues and fine-grained details, thereby improving robustness in complex scenes.

Although existing feature pyramid networks have demonstrated considerable value in object detection tasks, most of them are designed based on conventional natural scene images and exhibit limited adaptability to UAV-based aerial imagery. In aerial detection scenarios, targets typically exhibit small sizes, high object density, complex backgrounds, and significant scale variations, which introduce additional challenges for accurate detection. While several methods have shown partial effectiveness in such tasks, they often lack a comprehensive consideration of the global semantic information inherent in aerial imagery, thereby constraining overall detection performance.

2.3. Detection and attention-driven vision models

In the field of general object detection, Ren et al. (2016) proposed Faster R-CNN, which introduced the Region Proposal Network (RPN) to generate candidate regions directly from convolutional features. This innovation replaced the traditional selective-search algorithm and significantly improved detection efficiency. Moreover, the RPN and the feature extraction backbone share parameters, enabling an end-to-end training framework. Unlike the two-stage methods that perform detection in separate proposal and classification steps, one-stage detectors adopt a unified regression-based approach that predicts object locations and categories in a single forward pass, offering substantially higher speed with acceptable accuracy. Although two-stage detectors tend to achieve higher precision, they often suffer from slow inference and optimization complexity. Representative one-stage models include the YOLO series (Xue et al., 2024), SSD (Single Shot Multi-box Detector) (Liu et al., 2016), and RetinaNet (Ross and Dollár, 2017), which directly predict bounding boxes and class scores through a single network.

Building upon this paradigm, Vision Transformers (ViT) (Dosovitskiy et al., 2020) were initially designed for image classification,

and Facebook further extended this concept to propose DETR (Carion et al., 2020), redefining object detection as an end-to-end set prediction problem. DETR eliminates the need for region proposals, classification heads, and non-maximum suppression (NMS), offering a new paradigm that excels in capturing global context and handling complex multi-object scenes. However, its slow convergence and inference speed have motivated several subsequent improvements. Deformable-DETR (Zhu et al., 2020) introduced a deformable attention mechanism to selectively focus on informative regions, enhancing feature interaction efficiency. Efficient-DETR (Yao et al., 2021) optimized multi-scale feature extraction and decoder design to achieve comparable accuracy with significantly improved speed and computational efficiency. DN-DETR (Li et al., 2022b) reformulated the decoder queries as bounding box coordinates and incorporated noisy ground-truth boxes to accelerate convergence via a denoising strategy. Despite these advances, DETR-based detectors still demand substantial computational resources and extended training time, limiting their real-time applicability.

Beyond conventional object detection, recent research has extended attention mechanisms to video saliency detection and visual perception modeling. Zhang et al. (2025d) analyzed the limitations of state-of-the-art video saliency models and proposed the Single-feature Enhancement and Temporal Recurrence Network (SETR-Net), which introduces a Single-feature Channel Enhancement (SCE) module and a temporal recurrence mechanism to achieve more accurate fixation prediction across video sequences. Leveraging saliency-guided representation, Zhang et al. (2025h) proposed a dual-branch spatiotemporal network for driver attention prediction, integrating spatial and temporal cues through a feature-guided fusion module. Their model effectively captures static context and motion cues relevant to driver gaze, outperforming existing methods. Furthermore, saliency cues have been applied to video compression optimization. Zhang et al. (2025g) developed an efficient perceptual compression framework that integrates deep saliency prediction with the Just Noticeable Difference (JND) model, achieving higher compression ratios without sacrificing perceptual quality through adaptive bit allocation. In the domain of multimodal salient object detection, Zhang et al. (2025c) proposed the Asymmetric Light Perception and Progressive Decoding Network (ALPD-Net) for RGB-T detection. The model employs a pyramid structure and an Asymmetric Light-aware Interaction (ALI) module, where thermal features enhance RGB representations under low-light conditions, ensuring effective illumination adaptation. Finally, addressing occluded person re-identification (re-ID), Zhang et al. (2025e) introduced the Cross-Erasing Enhancement Network (CENet), which employs cross-guided feature suppression to mitigate the influence of occlusion, significantly improving Rank-1 accuracy under severe occlusion. These strategies of feature erasure and enhancement provide valuable insights for developing robust feature learning frameworks.

3. Methods

The overall architecture of GCRH is illustrated in Fig. 2, which consists of three main components: a backbone, a feature pyramid network, and a hybrid decoder with prediction heads. Specifically, multi-scale features from the last four stages of the backbone — denoted as $[S_2 \text{ (Optional)}, S_3, S_4, S_5]$ — are selected as input and fed into the Global-Aware Feature Pyramid Network (GAFPNN) to transform them into a set of key image features. To improve detection speed while maintaining accuracy, only the lowest-resolution feature map extracted by the backbone is processed through an encoder layer, while higher-resolution features directly participate in the subsequent feature fusion. This strategy significantly reduces the number of encoder layers, accelerating both training and inference. Subsequently, a query selection mechanism is employed to determine a fixed number of image features to serve as the initial object queries for the decoder. Finally, the hybrid decoder, equipped with prediction heads, iteratively refines these object queries to produce class labels and bounding boxes.

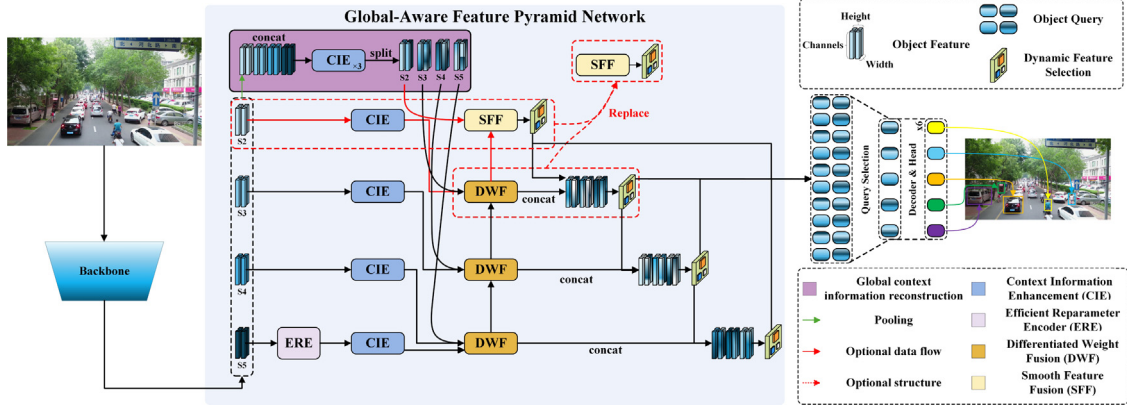


Fig. 2. The GCRH is a configurable hybrid detector designed for small object detection in UAV imagery. It consists of three core modules: (i) the Efficient Reparameter Encoder (ERE), which achieves global–local feature interaction under low computational cost through the Linear Additive Attention (LAA) mechanism and reparameterized batch normalization; (ii) the Global-Aware Feature Pyramid Network (GAFP), which reconstructs and propagates semantic contextual information across multi-scale feature layers; and (iii) an optional high-resolution S2 feature branch, enabling dynamic balance between accuracy and efficiency for small object representation. For clarity, the detailed architectures of submodules such as the Context Information Enhancement (CIE) and Differential Weight Fusion (DWF) are presented in subsequent sections.

3.1. Efficient reparameter encoder

Conventional Transformer encoders are primarily composed of multi-head self-attention mechanisms, which rely on full dot-product operations over token correlation matrices, resulting in extremely high computational complexity. In this work, inspired by the design principles of SwiftFormer (Shaker et al., 2023), we propose a novel Transformer encoder that integrates an efficient Linear Additive Attention (LAA) mechanism and Reparameterized Batch Normalization (RepBN) (Guo et al., 2024). The proposed Efficient Reparameter Encoder (ERE) combines the advantages of both techniques and incorporates a residual connection that adds the query directly to the attention output. This design not only enhances the retention of original features but also improves feature stability during training. The structural diagram of the Efficient Reparameter Encoder is illustrated in Fig. 3.

In conventional multi-head attention mechanisms, the relationship between each token and all other tokens is computed exhaustively, resulting in a computational complexity of $O(N^2)$, which is prohibitively expensive for large-scale inputs. Moreover, this global interaction scheme does not account for the contextual importance or influence of each token, potentially leading to the overemphasis of redundant information or the neglect of critical features. To address these issues, the proposed Linear Additive Attention (LAA) encodes queries and keys using linear projection layers. An adaptive scaling factor g is introduced to reweight each query before it interacts with the corresponding key. This allows the model to dynamically adjust the attention strength of each query, thereby establishing more effective global and local dependencies among tokens. In addition, LAA eliminates the need for a full token-to-token correlation matrix, avoiding costly full dot-product operations and substantially reducing the overall computational burden. The algorithmic flow of LAA is outlined as follows:

In conventional encoder architectures, Layer Normalization (LayerNorm) is commonly employed for feature normalization. However, LayerNorm requires statistical computation during both training and inference phases, which limits computational efficiency. To address this issue, we adopt Reparameterized Batch Normalization (RepBN) (Guo et al., 2024) as an alternative. To ensure stable training, we gradually replace LayerNorm with RepBN using a learnable weight factor γ , allowing a smooth transition throughout the training process. During training, RepBN preserves the original input as a residual term, enabling the model to learn more expressive and complex feature representations, thereby improving overall performance. In the inference

Algorithm 1: Linear Additive Attention

Input: Input $X \in \mathbb{R}^{B \times C \times H \times W}$, B: Batch Size, C: Channel, H: Height, W: Width

Require T: Transpose, Ln: L2Norm, w_g : Learnable scaling factor

Linear transformation $Q = \text{Linear}(X_{\text{flatten}}) \in \mathbb{R}^{B \times N \times D_q}$, $K = \text{Linear}(X_{\text{flatten}}) \in \mathbb{R}^{B \times N \times D_k}$

D_q, D_k : Dimensions of queries and keys, N: The length of the sequence in attention ($H \times W$)

foreach Tensor **do**

$Q_{Ln} = \text{Ln}(Q) = \frac{Q}{\|Q\|_2}$

Each scaling factor $g^{(i)} = Q_{Ln}^{(i)} \cdot w_g$, $w_g \in \mathbb{R}^{B \times N \times 1}$

Compute attention weight $A^{(i)} = \frac{Q_{Ln}^{(i)} \cdot K^{T(i)}}{D_q} \cdot g^{(i)}$

end

Attention weight: $G = \sum_{i=1}^{D_q} A^{(i)} \cdot Q_{Ln}^{(i)}$

Attention output: $O_{\text{weight}} = \text{Linear}(G \times K) + Q_{Ln}$

Final output: $O_{\text{final}} = \text{Linear}(O_{\text{weight}})$

$O_{\text{final}} \in \mathbb{R}^{B \times C \times H \times W}$

phase, RepBN is reparameterized into a standard Batch Normalization layer combined with a linear projection. This transformation introduces no additional parameters and significantly reduces inference complexity. The formula for RepBN is $\text{RepBN} = \text{BN}(X) + \eta X$, where η is a learnable parameter.

At the beginning of training, LayerNorm dominates, with $\gamma = 1$. As the training progresses, γ gradually decreases to 0, ensuring a smooth transition to RepBN. This process is described by the following formula:

$$\text{RepBN} = \gamma \text{LN}(X) + (1 - \gamma) \text{RepBN}(X) \quad (1)$$

$$\text{where } \gamma = \frac{S - S_{\text{current}}}{S} \quad (2)$$

where S represents the total number of training epochs and S_{current} denotes the current epoch.

3.2. Global-aware feature pyramid network

In UAV-based aerial image datasets, object detection across different categories is severely challenged by multi-scale feature variations, making it difficult for models to accurately recognize targets. This issue arises not only from the large scale discrepancies between different object types, but also from significant intra-class scale variation caused

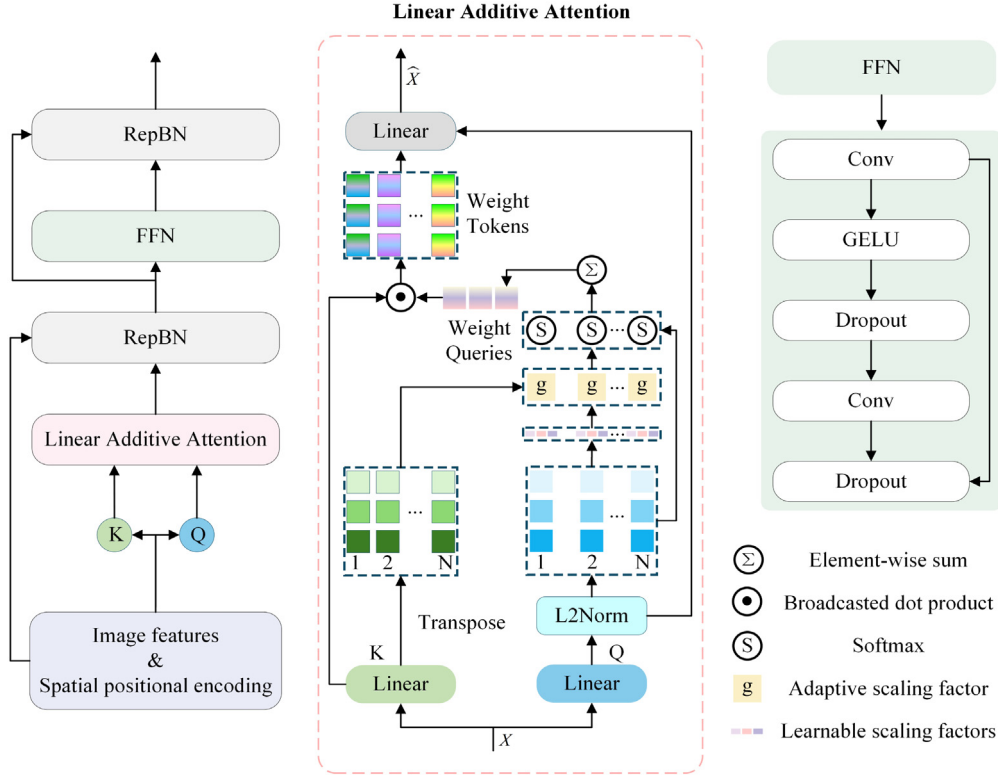


Fig. 3. Architecture of the Efficient Reparameterization Encoder (ERE), which improves global–local token interaction while reducing computational overhead by integrating linear additive attention and reparameterized batch normalization.

by diverse aerial viewing angles. To address the multi-scale challenge in aerial datasets, we propose a Global-Aware Feature Pyramid Network (GAFPN) to reconstruct global image context information. GAFPN is composed of three key feature processing components: (1) a Global Context Reconstruction Module, (2) a Differentiated Weighted Fusion Module and (3) a Dynamic Feature Selection Module. First, the multi-level features extracted from different stages of the backbone are integrated and fed into the Global Context Information Reconstruction (GCIR) module to recover both the missing contextual and critical feature information. These enriched features are then fused with the original backbone features through a Differential Weighted Fusion module, enabling comprehensive integration of high-level, mid-level, and low-level information. The top-down cascade transmission path injects abundant global semantics into the mid- and low-level feature maps, enhancing the model’s ability to perceive aerial image content. After each fusion stage, a Dynamic Feature Selection module is employed to refine the output features and improve their nonlinear representation capability.

Global context information reconstruction: The multi-scale features generated by the backbone network are used to extract both local and global contextual information. By applying downsampling operations, features from different levels are unified to the same scale and concatenated into a set of global features. These global feature sets are then fused using a context enhancement module, further improving feature representation and semantic awareness. Finally, the hierarchical feature sets are split and restored to the original input dimensions of each layer, providing rich semantic inputs for the subsequent feature pyramid. Through efficient downsampling strategies and a lightweight design (using depthwise convolution), the global context information reconstruction module significantly reduces computational costs while improving the efficiency of multi-scale feature fusion, enabling the model to better handle multi-scale objects and complex contextual relationships, thus enhancing the performance of the object detection task.

Context Information Enhancement (CIE) serves as the core structure for global context information reconstruction. This module uses both global average pooling and global max pooling to process the input features. The combined use of both methods allows the model to retain the strongest response values in the feature map, emphasizing significant regions in the image and reducing the impact of noise, while also summarizing average values to capture global semantic information, minimizing information loss. The pooled feature representations are separately fed into depth-wise strip convolution blocks along the horizontal and vertical directions to capture long-range directional dependencies. This design enables the extraction of extended spatial structures and simultaneously models inter-channel correlations, thereby enhancing the overall feature representation capability. The depthwise convolution operates separately in the horizontal and vertical directions, using larger kernels to capture global context information in both directions. Its computational complexity is only half that of conventional convolutions. Fig. 4 illustrates the structure of this module.

Differential Weight Fusion: The Differential Weight Fusion Module consists of the Smooth Feature Fusion Block and the Flexible Interpolation Fusion Block, as illustrated in Fig. 5. Both blocks perform fusion on images at different scales, utilizing the complementary features from lower and higher layers to enhance the model’s expressive capability. In the Smooth Feature Fusion Block, the Swish activation function (Fatima and Pethe, 2021) is used, providing a smoother gradient flow, which makes the model’s output more “fluid”. Unlike hard activation functions that quickly constrain outputs to 0 or 1, Swish offers a more continuous transition, which helps preserve fine-grained features and improves small object detection performance. During multi-scale feature fusion, the feature maps processed by the Swish activation function retain more details and information, allowing features from different scales to be better coordinated during fusion. The primary role of the Flexible Interpolation Fusion Block is to adjust the scale between features of different layers. The combination

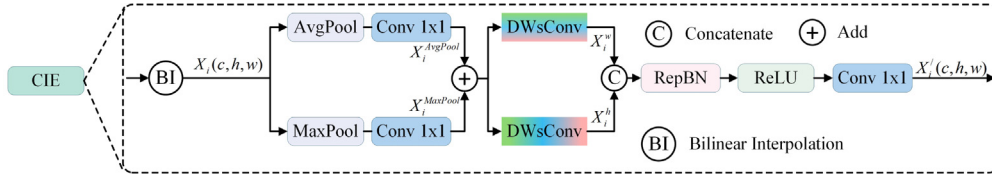


Fig. 4. Architecture of the Context Information Enhancement (CIE) module, which captures global contextual features using pooled features and directional strip convolutions.

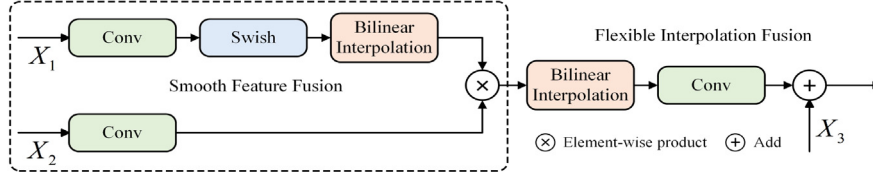


Fig. 5. Architecture of the Differentiated Weighted Fusion (DWF) module, consisting of a Smooth Feature Fusion (SFF) Block with Swish activation and a Flexible Interpolation Fusion (FIF) Block for multi-scale feature enhancement.

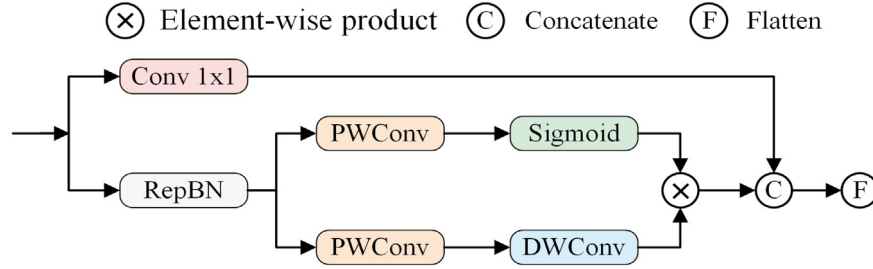


Fig. 6. Architecture of the Dynamic Feature Selection (DFS) module, which integrates cross-stage connections and a gating mechanism to refine and fuse feature maps at the final stage of the feature pyramid.

of bilinear interpolation and convolution optimizes the feature detail information, yielding better results. The lack of an activation function in the Flexible Interpolation Fusion Block reduces redundancy and lowers computational overhead.

Dynamic Feature Selection: The Dynamic Feature Selection module is constructed based on the design philosophy of CSPNet and a gating mechanism, and is applied to the final feature refinement stage of the feature pyramid, as shown in Fig. 6. To ensure data consistency of the features, the input feature X is normalized using RepBN, yielding X' . A pointwise convolution followed by a Sigmoid function generates the gating signal, $X_1 = \text{Sigmoid}(\text{PWConv}(X'))$. A separate branch of feature X' undergoes transformation using depthwise convolution and pointwise convolution, resulting in $X_2 = \text{DWConv}(\text{PWConv}(X'))$. The gating signal X_1 is then used to weight X_2 , producing X_3 . Finally, X_3 is concatenated with $\text{Conv}_{1 \times 1}(X)$ (after adjusting the channel number) to complete the feature refinement process. The dense block employs PWConv and DWConv as the primary computational structures, maximizing the reduction in parameter count and computational complexity, thus accelerating both the training and inference phases of the model. Cross-order connections help the gradient propagate more easily, preserve the input features after the feature transformation, and enhance the efficiency of information flow while introducing a degree of non-linearity.

3.3. Configurable multi scale design

To meet the demands of both accuracy and inference speed in practical applications, most state-of-the-art object detectors are typically available in multiple model scales. Following this design principle, the proposed GCRH framework also emphasizes model scalability and adaptability. Specifically, GCRH offers two model variants: a large-scale version (GCRH-L), which incorporates the shallow feature layer

S_2 from the backbone into the subsequent feature fusion process, and a lightweight version (GCRH-S), which omits this layer for improved efficiency.

The S_2 feature layer is located at an early stage of the backbone and retains higher-resolution features with richer spatial details. These shallow features are particularly critical for identifying small objects in complex aerial scenes, as they complement deeper contextual semantics and contribute to more complete fused representations. However, incorporating high-resolution feature maps increases computational overhead and inference latency. To balance detection accuracy and computational cost, GCRH enables users to select an appropriate model variant based on deployment constraints. GCRH-L is recommended for accuracy-critical scenarios with sufficient computational resources, while GCRH-S offers lower latency and controlled performance degradation, making it suitable for real-time applications.

Moreover, GCRH is designed to be modular and compatible with a wide range of backbone networks, from lightweight architectures such as ShuffleNet and MobileNet to more powerful ones like ResNet-101 and CSPDarkNet. In summary, by flexibly integrating the S_2 layer and supporting backbone modularity, GCRH establishes a highly scalable detection framework that achieves a favorable trade-off between accuracy and efficiency across various deployment scenarios.

4. Experiments

4.1. Datasets

GCRH is evaluated on two challenging benchmark datasets — VisDrone2019 and TinyPerson — both of which are specifically designed for small object detection in UAV-based aerial imagery. VisDrone2019 is one of the most widely adopted UAV detection datasets. It comprises images and videos captured by drone-mounted cameras

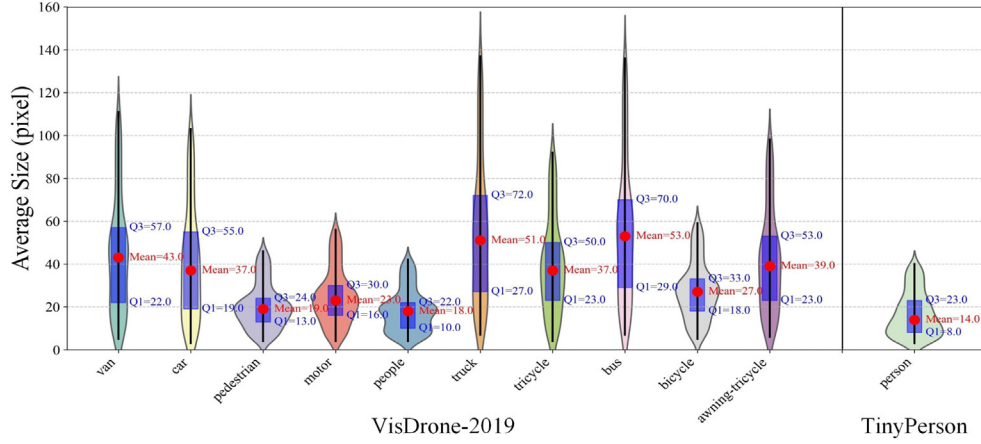


Fig. 7. Violin plot of scale distribution for VisDrone2019-DET and TinyPerson.

across 14 cities in China, covering both urban and rural environments. The dataset includes 10 object categories and exhibits a wide range of scene densities, from sparse to highly crowded. In total, it contains 7019 images, with 6471 for training, 548 for validation, and 1610 for testing. TinyPerson (Yu et al., 2020) is a dataset tailored for detecting extremely small pedestrian instances in real-world scenarios. These targets occupy a very small proportion of the image, often appear blurry, and are highly susceptible to background interference due to their lack of clear visual structure. The dataset contains 1610 images, with 794 used for training and 816 for testing. In our experiments, the “sea person” and “earth person” categories are merged into a single “person” class. To better characterize object scale, TinyPerson divides instances into six categories based on object area: tiny1 (area $\leq 8 \times 8$), tiny2 ($8 \times 8 < \text{area} \leq 12 \times 12$), tiny3 ($12 \times 12 < \text{area} \leq 20 \times 20$), tiny (area $\leq 20 \times 20$, including tiny1-3), small ($20 \times 20 < \text{area} \leq 32 \times 32$), and non-small (area $> 32 \times 32$). As shown in Fig. 7, both datasets contain a high proportion of small objects (less than 30 px) and exhibit large-scale variation. These observations confirm that the top-down perspective of UAV imaging results in significant differences in object scale and spatial distribution compared to natural scene images. Consequently, object detection performance in aerial imagery tends to be inferior to conventional settings, imposing higher demands on model adaptability and robustness.

4.2. Implementation details

All experiments were implemented using the PyTorch deep learning framework based on the MMDetection and Ultralytics toolboxes. The proposed GCRH framework was trained and evaluated on two NVIDIA Tesla V100 GPUs. The input image size of all detectors was uniformly set to $640 \times 640 \times 3$, and the batch size was fixed at 16. To ensure experimental fairness and validity, no manual parameter tuning was performed across models. All methods were trained and evaluated under the same hardware environment to eliminate potential performance bias caused by hardware discrepancies. The model scales were determined according to the number of parameters. Evaluation is conducted on two benchmark test sets: the VisDrone2019 test set with 10 object categories and the TinyPerson test set with a single “person” category. The COCO evaluation protocol is adopted as the primary performance metric. For the VisDrone2019 dataset, both ResNet-18 and ResNet-50 are used as backbone feature extractors. Given the higher difficulty of the TinyPerson dataset, ResNet-50 is selected exclusively to ensure stable and reliable results. Regarding the training details, all experiments on the VisDrone dataset followed a 500-epoch schedule with AdamW optimizer (initial learning rate $= 1 \times 10^{-4}$, weight decay $= 0.05$). The learning rate was decayed using a cosine annealing schedule. We applied multi-scale training (random resizing between 480–672

with a step of 32), Mosaic and MixUp augmentations, color jitter, and random horizontal flipping ($p = 0.5$). These augmentations and regularization strategies aim to enhance the model’s robustness to scale variation and background clutter, which are particularly prevalent in UAV-based imagery. This training protocol ensures that differences in performance originate from architectural innovations rather than hyperparameter discrepancies. The reported results thus objectively reflect the improvements brought by the proposed modules.

4.3. Comparison with sota detectors

A comprehensive comparison is conducted between the proposed method and several state-of-the-art object detection models, with a particular focus on evaluating GCRH in terms of accuracy, inference speed, and multi-scale detection performance. As summarized in Tables 1 and 2, GCRH achieves competitive results across all evaluation dimensions, demonstrating superior detection accuracy and faster inference speed. Notably, GCRH exhibits significant advantages in real-time and efficiency-critical scenarios.

VisDrone-2019: From the accuracy metrics, GCRH demonstrates outstanding performance across different evaluation criteria. The GCRH model with the R-50 backbone achieves an AP_{50} of 47.0, surpassing similar state-of-the-art models such as RT-DETR-R50 (38.5), DINO-DETR-R50 (43.4), DDQ-DETR-R50 (46.3) and PSWP-DETR-R50 (43.2). Additionally, on the more challenging AP_{75} metric, GCRH-R50 reaches 27.4, surpassing DDQ-DETR-R50 (27.2), while also maintaining a significant advantage in inference speed. The GCRH with a ResNet-50 backbone achieves an AP_S of 16.4%, which significantly surpasses the classical one-stage detectors YOLOv10-X (11.5%) and ATSS (14.3%), and performs on par with or better than the two-stage frameworks Faster R-CNN (15.6%) and Cascade R-CNN (9.8%). Moreover, GCRH also outperforms transformer-based architectures such as PSWP-DETR (15.8%) and DDQ-DETR (16.2%), demonstrating that the proposed global-context reparameterization mechanism effectively enhances the extraction of fine-grained features for small objects. Although GCRH achieves $AP_M = 39.0\%$ and $AP_L = 54.0\%$, showing only marginal improvement over DDQ-DETR-R50 ($AP_M: 38.9\%$, $AP_L: 53.6\%$), it still maintains a clear advantage compared to other state-of-the-art models such as YOLOv5-L (40.7%), ATSS-R50 (40.9%), RetinaNet-R50 (42.3%), TOOD-R50 (43.0%), Faster-RCNN-R50 (43.0%), RT-DETR-R50 (43.4%), YOLOv8-L (45.6%), YOLOv10-X (46.6%), DMTNet (42.5%), and LMSOE-Net-L (45.6%). These scale-specific results collectively verify the superiority and robustness of GCRH in multi-scale detection tasks. In particular, for dense small-object and low-resolution UAV imagery, GCRH significantly improves detection accuracy while maintaining competitive performance for medium and large objects. When using the R-18 backbone, the GCRH model outperforms models

Table 1

Comparison of GCRH with state-of-the-art object detectors on the VisDrone2019 dataset in terms of accuracy and computational efficiency.

Model	Backbone	AP ₅₀	AP ₇₅	AP _S	AP _M	AP _L	GFLOPs	Params (M)	FPS _{bs=1}
YOLOv5-M (Jocher et al., 2022)	–	30.1	18.3	7.7	27.6	38.3	64	21.2	63
YOLOv8-M (Xue et al., 2024)	–	34.7	20.4	9.8	30.7	42.7	78	25.8	85
YOLOv10-L (Wang et al., 2024)	–	35.8	20.7	10.1	30.8	43.4	120	24.4	126
ATSS (Zhang et al., 2020)	R-18	35.5	20.5	12.0	30.7	35.3	61	19.0	–
RetinaNet (Ross and Dollár, 2017)	R-18	34.3	19.2	9.7	30.9	37.9	169	19.8	–
TOOD (Feng et al., 2021)	R-18	29.5	18.3	8.1	26.1	37.2	144	18.9	49
Faster-RCNN (Ren et al., 2016)	R-18	36.7	22.5	13.7	32.0	37.3	140	28.2	47
Cascade-rcnn (Cai and Vasconcelos, 2019)	R-18	35.1	20.7	9.4	30.5	35.9	168	48.4	–
DINO-DETR (Zhang et al., 2022)	R-18	38.1	22.9	13.7	36.0	38.1	200	25.3	–
DDQ-DETR (Zhang et al., 2023a)	R-18	38.9	23.7	14.0	37.2	38.4	206	55.6	–
RT-DETR (Zhao et al., 2024a)	R-18	36.5	20.9	11.6	31.0	31.3	61	20.1	136
DMTNet (Zhang et al., 2025b)	R-18	35.4	18.2	9.8	28.0	29.9	115	27.8	–
PSWP-DETR (Zhong et al., 2025)	R-18	37.1	21.8	13.2	33.1	32.5	63	41.2	68
LMSOE-Net-M (Ma et al., 2025)	–	30.5	18.4	6.9	28.9	32.1	65	17.5	151
GCRH (Ours)	R-18	39.1	23.9	14.2	37.4	38.8	58	24.2	140
YOLOv5-L (Jocher et al., 2022)	–	31.9	19.6	8.3	29.7	40.7	109	46.5	55
YOLOv8-L (Xue et al., 2024)	–	36.9	22.0	10.6	33.0	45.6	139	43.2	72
YOLOv10-X (Wang et al., 2024)	–	38.0	22.4	11.5	33.3	46.6	183	29.5	95
ATSS (Zhang et al., 2020)	R-50	40.2	24.6	14.3	35.7	40.9	114	31.9	–
RetinaNet (Ross and Dollár, 2017)	R-50	38.7	23.0	11.5	36.2	42.3	215	39.3	–
TOOD (Feng et al., 2021)	R-50	32.2	20.5	9.5	30.3	43.0	189	31.8	30
Faster-RCNN (Ren et al., 2016)	R-50	42.4	24.4	15.6	36.5	43.0	187	41.2	28
Cascade-RCNN (Cai and Vasconcelos, 2019)	R-50	32.7	21.1	9.8	31.1	41.8	215	69.3	–
DINO-DETR (Zhang et al., 2022)	R-50	43.4	24.7	14.6	37.5	50.8	249	47.5	–
DDQ-DETR (Zhang et al., 2023a)	R-50	46.3	27.2	16.2	38.9	53.6	254	67.5	–
RT-DETR (Zhao et al., 2024a)	R-50	38.5	22.4	12.7	32.5	43.4	136	42.3	98
DMTNet (Zhang et al., 2025b)	R-50	41.1	23.6	13.4	31.7	42.5	121	41.7	–
PSWP-DETR (Zhong et al., 2025)	R-50	43.2	25.1	15.8	35.3	44.4	140	97.9	34
LMSOE-Net-L (Ma et al., 2025)	–	42.2	24.3	10.0	33.9	45.6	134	29.1	110
GCRH (Ours)	R-50	47.0	27.4	16.4	39.0	54.0	128	46.8	101

Table 2

Comparison of GCRH with state-of-the-art object detectors on the TinyPerson dataset in terms of detection accuracy.

Model	AP ₂₅ ^{tiny}	AP ₅₀ ^{tiny}	AP ₇₅ ^{tiny}	AP ₅₀ ^{tiny1}	AP ₅₀ ^{tiny2}	AP ₅₀ ^{tiny3}	AP ₅₀ ^{small}
YOLOv5-L (Jocher et al., 2022)	62.9	38.1	2.5	13.6	38.9	47.2	52.1
YOLOv8-L (Xue et al., 2024)	64.1	40.1	3.8	27.4	42.7	50.8	55.7
YOLOv10-X (Wang et al., 2024)	69.7	48.0	4.9	33.2	49.6	63.0	63.4
ATSS (Zhang et al., 2020)	61.9	34.0	2.2	13.2	38.0	47.4	48.9
RetinaNet (Ross and Dollár, 2017)	62.6	34.1	2.3	13.4	38.8	47.5	48.7
TOOD (Feng et al., 2021)	60.8	33.0	2.0	12.7	37.0	46.8	47.9
Faster-RCNN (Ren et al., 2016)	69.3	47.9	5.7	31.5	52.6	60.0	65.2
Cascade-RCNN (Cai and Vasconcelos, 2019)	70.0	48.4	5.8	32.4	52.9	61.1	66.9
DINO-DETR (Zhang et al., 2022)	78.9	60.4	7.7	44.0	63.3	67.9	69.4
DDQ-DETR (Zhang et al., 2023a)	79.6	59.0	7.8	43.9	63.5	68.1	68.7
RT-DETR (Zhao et al., 2024a)	68.6	49.1	6.0	33.4	58.8	64.5	66.7
GCRH (Ours)	80.1	60.7	7.9	44.1	63.7	68.1	69.6

such as YOLOv5-M, YOLOv8-M, YOLOv10-L, ATSS-R-18, RetinaNet-R-18, TOOD-R-18, Faster-RCNN-R-18, Cascade-RCNN-R18, DINO-DETR-R-18, DDQ-DETR-R18, RT-DETR-R-18, DMTNet-R-18, PSWP-DETR-R-18 and LMSOE-Net-M with an increase of 29.9%/12.7%/9.2%/10.1%/13.9%/32.5%/6.5%/11.4%/2.6%/0.5%/7.1%/10.4%/5.4%/28.2% in AP₅₀. For more difficult metrics such as AP₇₅ and AP_S, GCRH still leads and achieves the best results, demonstrating that the model can maintain its performance with different backbone networks.

In terms of inference speed, the proposed GCRH demonstrates remarkable performance. With the ResNet-50 backbone, GCRH achieves a throughput of 101 FPS, significantly outperforming high-accuracy models such as DINO-DETR and DDQ-DETR. Compared to representative real-time detectors from the YOLO series, GCRH also exhibits competitive speed, with FPS improvements of 83.6% and 40.3% over YOLOv5 and YOLOv8, respectively. Although there is a slight gap in FPS compared to the fastest model LMSOE-Net-L, GCRH maintains a substantial advantage in detection accuracy, exceeding it by 28.19%. This trade-off is considered acceptable in real-world UAV detection scenarios, where both accuracy and responsiveness are critical.

In terms of computational efficiency, GCRH maintains a leading position when equipped with the ResNet-50 backbone. It achieves a GFLOPs of 128, which is significantly lower than many mainstream

detectors. Specifically, GCRH reduces FLOPs by 49.6% compared to DDQ-DETR (254), 48.6% compared to DINO-DETR (249), 31.5% compared to Faster R-CNN (187), 32.3% compared to TOOD (189), and 40.5% compared to RetinaNet (215). When compared with more efficient models such as YOLOv5-L (109), RT-DETR (136), YOLOv8-L (139) and PSWP-DETR-R50(140), GCRH demonstrates comparable computational cost while achieving significantly higher accuracy, with AP₅₀ improvements of 47.3%, 22.1%, 27.4% and 8.79% respectively. Using the lightweight ResNet-18 backbone, GCRH further reduces the computational burden to 58 GFLOPs, which is 4.9% lower than ATSS (61) and RT-DETR (61), and 10.3% and 34.5% lower than YOLOv5-M (64) and YOLOv8-M (78), respectively. These results confirm that GCRH achieves the highest AP scores across different backbone configurations while maintaining low computational complexity, demonstrating its effectiveness in balancing accuracy and efficiency.

In terms of parameter scale, GCRH-R50 has 46.8M parameters, which is comparable to YOLOv5-L (46.5M) and DINO-DETR-R50 (47.2M). However, GCRH demonstrates a significant advantage in both detection speed and accuracy. When comparing GCRH-R18 (24.2M) to models with similar parameter sizes, such as YOLOv8-M (25.8M), YOLOv10-L (24.4M) and Faster-RCNN-R18 (28.2M), GCRH outperforms them comprehensively in both detection accuracy and speed. Fig.



Fig. 8. The visualization of detection results for different models, with regions showing significant differences highlighted using red bounding boxes. Some enlarged images of the red bounding boxes are located in the corners.

8 illustrates the visual detection results of GCRH under various levels of scene congestion and different shooting angles.

TinyPerson: To validate the effectiveness of the proposed GCRH detector, we conduct a comprehensive comparison with various state-of-the-art object detection methods on the TinyPerson benchmark dataset. Table 2 reports the average precision (AP) under different evaluation settings, including AP_{25}^{tiny} , AP_{50}^{tiny} , AP_{75}^{tiny} , $AP_{50}^{tiny1-3}$, and AP_{50}^{small} .

Experimental results demonstrate that the proposed method outperforms existing state-of-the-art detectors across all evaluation metrics. Specifically, GCRH achieves 80.1%, 60.7%, and 7.9% on AP_{25}^{tiny} , AP_{50}^{tiny} , and AP_{75}^{tiny} , respectively, surpassing strong baselines such as DINO-DETR (78.9%, 60.4%, 7.7%) and DDQ-DETR (79.6%, 59.0%, 7.8%). These results highlight the superior robustness and localization accuracy of GCRH under high IoU thresholds. Moreover, GCRH also achieves leading performance on AP_{50}^{tiny1} (44.1%), AP_{50}^{tiny2} (63.7%), and AP_{50}^{tiny3} (68.1%), significantly outperforming classic detectors such as YOLOv5-L, Faster R-CNN, and YOLOv10-X, thereby demonstrating its stronger capability in detecting ultra-small and low-resolution objects. Importantly, GCRH achieves the best result of 69.6% on AP_{50}^{small} , further validating its advantage in small object detection and its applicability to UAV aerial imagery and other real-world scenarios. These findings clearly demonstrate the effectiveness of the proposed global semantic reconstruction strategy and cross-level semantic alignment mechanism in enhancing small object perception and representation. Fig. 9 presents a comparison of AP_{50} performance across 11 object categories between GCRH models with two backbone variants and other baseline detectors on both datasets.

4.4. Comparison with other feature pyramid networks

To comprehensively evaluate the performance of the Global-Aware Feature Pyramid Network (GAFPN) in object detection tasks, comparative experiments were conducted on the VisDrone2019 test set, where GAFPN was compared with several other advanced feature pyramid networks based on RT-DETR. The focus of the analysis was on the performance of GAFPN. The experimental results are shown in Table 3.

When using ResNet-18 as the backbone, GAFPN achieved an AP_{50} of 37.5%, significantly surpassing standard FPN (35.4%), SlimNeck (36.4%), AugFPN (36.5%), and BiFPN (36.9%). On the more challenging AP_{75} metric, GAFPN reached 22.3%, outperforming other models. With ResNet-50 as the backbone, GAFPN achieved an AP_{50} of 41.1%, surpassing SlimNeck (38.6%), BiFPN (39.0%), and SSFPN (39.9%), delivering the best performance among all compared models. GAFPN also achieved an AP_{75} of 23.4%, once again outperforming other models such as CFPT (23.0%) and AugFPN (22.5%), showcasing its superior high-precision detection capability. Small object detection in aerial images is a challenging task in object detection. With ResNet-18 and ResNet-50 as the backbones, GAFPN achieved AP_s of 12.6% and 13.9%, respectively, setting the best results among the compared models, demonstrating higher sensitivity to small objects in aerial images. In the medium object detection task, with ResNet-18 as the backbone, GAFPN (33.2%) outperformed most models such as FPN (30.4%), BiFPN (31.7%), and SlimNeck (32.0%). With ResNet-50 as the backbone, GAFPN's AP_{50} reached 36.0%, significantly surpassing the baseline FPN (31.8%) and outperforming other advanced feature pyramid structures. In large object detection tasks, with ResNet-18

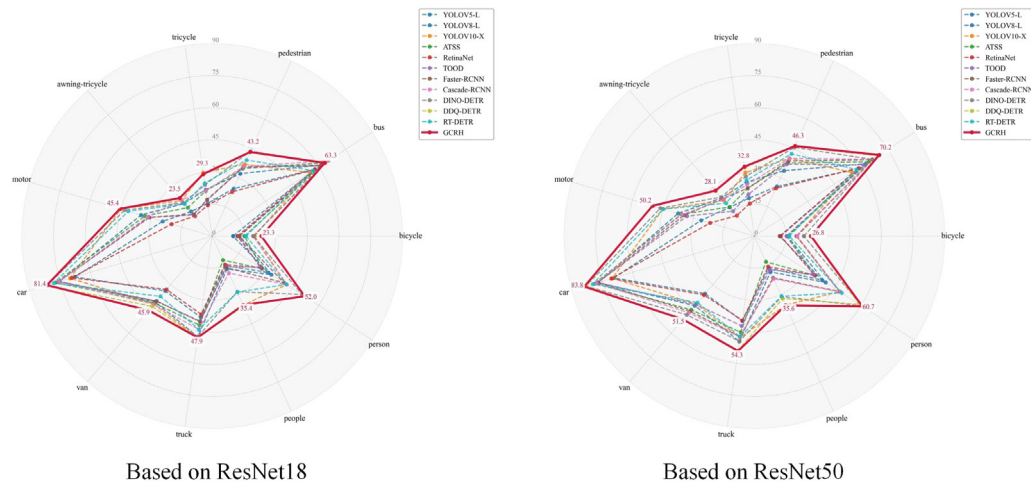


Fig. 9. The radar chart of AP_{50} for each object category across different models. The larger the area enclosed by the outer lines, the better the performance of the corresponding method.

Table 3

Comparison of GAFFN with state-of-the-art feature pyramid networks on the VisDrone2019 dataset using RT-DETR with different backbone networks.

Method	Backbone	Params (M)	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
RT-DETR (Zhao et al., 2024a)	R-18	20.1	36.5	20.9	11.6	31.0	31.3
FPN (Lin et al., 2017)	R-18	20.7	35.4	18.8	10.8	30.4	31.1
PAFPN (Liu et al., 2018)	R-18	23.3	36.1	19.7	9.9	31.0	31.7
BiFPN (Tan et al., 2020)	R-18	21.9	36.9	20.9	11.5	31.7	34.2
AugFPN (Guo et al., 2020)	R-18	21.5	36.5	18.5	10.4	29.9	33.0
FPG (Chen et al., 2020)	R-18	56.7	32.9	18.3	9.7	29.8	35.6
DRFPN (Ma and Chen, 2020)	R-18	24.9	33.5	18.4	9.2	30.0	38.3
RCFPN (Zong et al., 2021)	R-18	24.4	36.8	21.5	11.1	31.4	33.6
SSFPN (Hong et al., 2021)	R-18	25.3	37.2	21.7	12.1	32.7	35.2
SlimNeck (Li et al., 2024)	R-18	21.1	36.4	21.0	11.9	32.0	31.4
AFPN (Yang et al., 2023)	R-18	19.7	37.1	21.0	11.7	31.8	32.8
CFPT (Du et al., 2024)	R-18	22.0	37.3	22.1	12.2	33.1	35.4
GAFFN (Ours)	R-18	24.4	37.5	22.3	12.6	33.2	38.4
RT-DETR (Zhao et al., 2024a)	R-50	42.3	38.5	22.4	12.7	32.5	43.4
FPN (Lin et al., 2017)	R-50	37.7	38.1	21.8	10.9	31.8	42.7
PAFPN (Liu et al., 2018)	R-50	40.3	38.4	22.1	11.1	32.1	43.6
BiFPN (Tan et al., 2020)	R-50	37.4	39.0	22.4	12.2	33.3	44.5
AugFPN (Guo et al., 2020)	R-50	39.5	38.7	22.5	12.4	32.6	44.7
FPG (Chen et al., 2020)	R-50	71.0	37.5	22.0	11.3	35.4	38.9
DRFPN (Ma and Chen, 2020)	R-50	41.3	36.2	22.1	11.0	35.2	39.4
RCFPN (Zong et al., 2021)	R-50	37.6	39.1	22.3	11.9	32.7	44.9
SSFPN (Hong et al., 2021)	R-50	42.4	39.9	23.1	13.3	35.4	45.3
SlimNeck (Li et al., 2024)	R-50	38.1	38.6	22.0	12.8	34.2	43.9
AFPN (Yang et al., 2023)	R-50	45.3	39.2	22.2	13.0	33.9	44.4
CFPT (Du et al., 2024)	R-50	38.9	40.7	23.0	13.6	35.9	45.2
GAFFN (Ours)	R-50	44.9	41.1	23.4	13.9	36.0	46.1

as the backbone, GAFFN achieved an AP_L of 38.4%, significantly outperforming SlimNeck (31.4%), AFPN (32.8%), and RCFPN (33.6%). With ResNet-50 as the backbone, GAFFN achieved an AP_L of 46.1%, far surpassing SlimNeck (43.9%), AFPN (44.4%), AugFPN (44.7%), RCFPN (44.9%), and CFPT (45.2%), demonstrating its strong capability in large object detection. Through the above comparative analysis, GAFFN excels in major metrics (AP_{50} , AP_{75} , AP_S , AP_M , AP_L), especially in small object detection and high-precision detection, showing excellent multi-scale adaptability. Although its parameter count is slightly higher than that of standard FPN, its performance improvement is substantial, demonstrating better parameter utilization efficiency. Overall, GAFFN achieves a good balance between detection performance and computational complexity through efficient feature fusion and optimized design, providing strong support for high-precision object detection tasks in aerial images.

4.5. Analysis of model limitations and failure cases in complex UAV scenarios

To further validate the effectiveness of the proposed GCRH, we conducted a qualitative comparison with several representative methods from recent years, as illustrated in Fig. 10. The experiments were carried out on the VisDrone2019 and TinyPerson datasets, focusing on dense small-object overlap, illumination interference, and complex nighttime scenarios. Based on the characteristics of these datasets, we selected representative images that met the qualitative analysis criteria and visualized the prediction results using heatmaps to intuitively demonstrate detection performance.

Under dense small-object overlap conditions in the TinyPerson dataset (first row), all models exhibited missed detections when handling extremely small objects that shared high texture similarity with

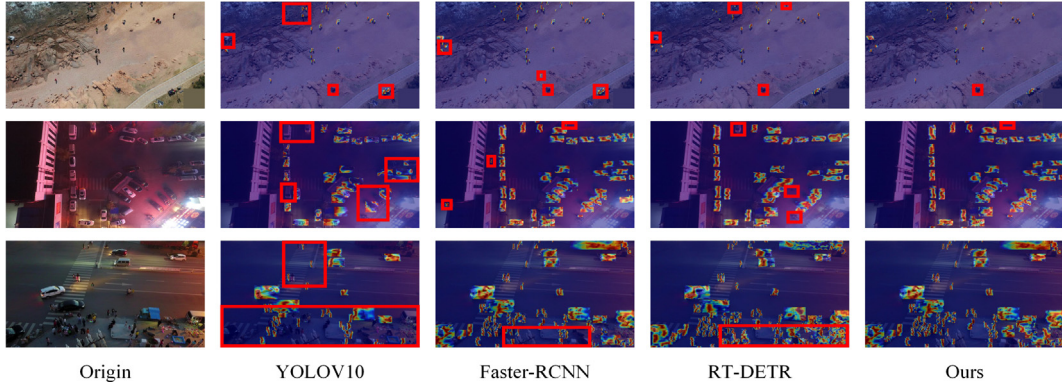


Fig. 10. Qualitative visualization of detection performance in complex UAV scenes.

the background. Both one-stage and two-stage detectors failed to accurately identify the number of overlapping objects in the lower-right region of the image, leading to false predictions. In contrast, DETR-based architectures did not exhibit such deficiencies, showing more stable responses to object occlusion.

In the illumination interference condition of the VisDrone2019 dataset (second row), the models revealed significant performance differences. The one-stage detectors showed clear limitations, losing a large number of critical objects and demonstrating limited focus across object categories. Two-stage and DETR-based models achieved better overall results but still suffered from missing and misclassified small objects, especially in the upper-right region. In complex nighttime scenes (third row), our GCRH achieved the best performance among all compared models, successfully distinguishing object categories and quantities with clear visual boundaries. While one-stage models failed to capture most key targets, Faster R-CNN and RT-DETR detected the majority but produced localization errors and false positives when multiple small objects overlapped in the lower-right area.

Overall, under extreme and cluttered conditions, the superior performance of our model can be attributed to the efficient re-parameterized encoder, which preserves fine-grained feature representations and effectively captures multi-scale contextual cues. However, we also observed that in certain extreme scenarios — particularly those involving excessive background noise, severe illumination imbalance, or motion blur — the model occasionally generated false positives in regions with strong light reflection or highly textured backgrounds. These errors are likely caused by local feature over-enhancement and imprecise attention allocation during the feature fusion process, which may slightly distort the semantic boundaries of adjacent small objects. Future improvements will aim to incorporate adaptive illumination normalization and spatial-aware attention modulation to further mitigate these limitations.

4.6. Ablation study

To further investigate the contribution of each proposed component, we progressively integrated the designed modules into the baseline model (RT-DETR with a ResNet-50 backbone and without FPN). The results on the VisDrone 2019 and TinyPerson datasets are summarized in Table 4. Introducing the Global Context Information Reconstruction (GCIR) module improved the baseline's performance by +3.62% AP_{50}^{tiny} , +6.23% AP_{50} , +5.36% AP_{75} , and +6.29% AP_S , demonstrating that the reconstruction of long-range contextual cues effectively enhances global awareness. The computational overhead was marginal, with parameters increasing by only 0.7 M and GFLOPs by 4, while the inference speed dropped slightly (−5 FPS). After combining GCIR with the Global-Aware Feature Pyramid Network (GAFPN), the detection accuracy was further boosted by +7.43% AP_{50}^{tiny} and +11.95% AP_{50}

compared with the baseline, confirming that multi-scale semantic alignment strengthens feature consistency across scales. When replacing the standard multi-head self-attention mechanism with the proposed Efficient Re-parameterized Encoder (ERE), the model achieved noticeable computational gains while maintaining accuracy improvements. Specifically, compared with the baseline, the parameter count was reduced by 3.9%, GFLOPs decreased by 13.1%, and inference speed increased by 8.2% (row 3 vs. row 1). Under the same GCIR + GAFPN configuration, the inclusion of ERE (row 5 vs. row 4) further reduced parameters by 3.1%, GFLOPs by 6.7%, and improved FPS by 11.1%. These results quantitatively demonstrate that ERE effectively balances accuracy and efficiency through lightweight linear token interaction.

In addition, the two model variants of GCRH mainly differ in whether the high-resolution S_2 layer is included for small-object representation. As shown in Table 4, introducing the S_2 feature layer significantly enhances detection accuracy, raising AP_{50}^{tiny} , AP_{50} , AP_{75} , and AP_S to 55.9%, 47.3%, 27.2%, and 15.9%, respectively. These represent gains of +13.8%, +22.9%, +21.4%, and +25.2% over the baseline. However, this accuracy gain comes at a cost: the parameter count and GFLOPs increase by 13.1% and 22.4%, while the FPS decreases by 26.7%. Hence, S_2 acts as a double-edged sword, providing a flexible option between high-precision detection and real-time inference depending on the application scenario. Fig. 11 presents the heatmaps generated by the prediction heads of various ablation models under different camera viewpoints across the two datasets.

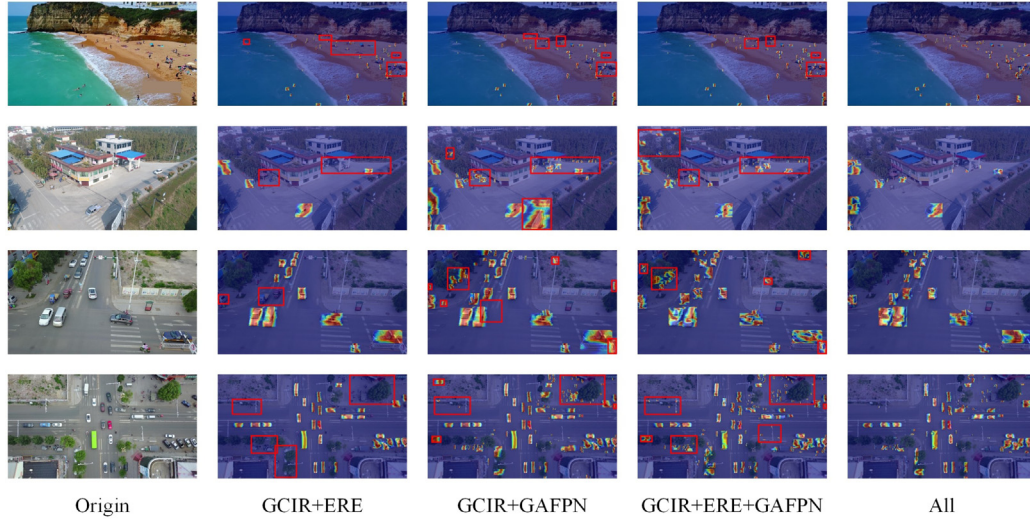
5. Conclusion and discussion

This paper presents GCRH, a Global Context Reconstruction Hybrid detector tailored for object detection tasks in UAV aerial imagery. The proposed framework addresses key challenges such as the high proportion of small objects, insufficient contextual information, and poor retention of critical features. GCRH integrates two core modules: an Efficient Reparameterization Encoder (ERE) and a Global-aware Feature Pyramid Network (GAFPN). The ERE mitigates the heavy computational burden and interaction redundancy commonly encountered in traditional multi-head self-attention encoders. Meanwhile, the GAFPN reconstructs and distributes global contextual semantics, while incorporating a gated dynamic feature selection mechanism to suppress redundant interference and enhance the expressiveness of fused features. To accommodate diverse deployment requirements, GCRH adopts a configurable architecture: the GCRH-L variant introduces a high-resolution S_2 feature layer for small object detection into the backbone, significantly improving detection precision, whereas GCRH-S omits this layer to optimize inference speed. Extensive experiments on VisDrone2019 and TinyPerson demonstrate the superiority of the proposed GCRH. Results show that ERE significantly reduces parameters and FLOPs while maintaining competitive accuracy, GAFPN effectively

Table 4

Ablation study of key performance metrics on VisDrone-2019 and TinyPerson datasets.

GCIR	ERE	GAFFN	S_2 layer	AP_{50}^{tiny}	AP_{50} (%)	AP_{75} (%)	AP_s (%)	Params (M)	GFLOPs	FPS _{bs=1}
Baseline				49.1	38.5	22.4	12.7	42.3	136	98
✓	×	×	×	50.8	40.9	23.6	13.5	43.0	140	93
×	✓	×	×	49.4	38.7	22.5	12.9	40.7	122	106
✓	×	✓	×	52.7	43.1	24.5	14.2	46.6	149	81
✓	✓	✓	×	52.5	42.9	24.4	14.1	45.2	134	90
✓	✓	✓	✓	55.9	47.3	27.2	15.9	51.1	164	71

**Fig. 11.** Visualization of ablation experiments. Areas with significant differences are marked with red boxes.

enhances cross-level semantic consistency, and the configurable design provides a controllable trade-off between detection precision and computational efficiency.

It is worth emphasizing that all modules within the proposed GCRH framework exhibit model-independent and modular characteristics, allowing them to be seamlessly integrated into diverse backbone networks, neck structures, and detection architectures without requiring structural modification. This design flexibility endows GCRH with the potential to serve as a set of universal detection components, adaptable to various vision-based applications. In particular, the lightweight design and plug-and-play properties of the Efficient Re-parameterized Encoder (ERE) and Global-Aware Feature Pyramid Network (GAFFN) enable straightforward deployment across both transformer-based and convolutional detection frameworks. The Configurable Multi-Scale Design further enhances adaptability by providing a tunable balance between detection accuracy and computational efficiency. Overall, GCRH achieves a favorable balance between accuracy and speed, making it suitable for deployment in UAV-based perception systems.

In future work, the proposed GCRH framework can be further enhanced in terms of robustness and adaptability through several promising directions. First, incorporating hierarchical or multi-stage detection strategies — such as global-to-local or coarse-to-fine paradigms — may lead to more accurate object localization in densely populated scenes. Second, combining model compression techniques such as neural architecture search (NAS), quantization, and pruning with hardware-aware optimization can effectively reduce model size and further improve computational efficiency. Third, the design of context-aware data augmentation or adversarial sample generation strategies may help improve generalization for small, occluded, or low-resolution objects. Moreover, integrating multimodal inputs (e.g., depth maps or thermal infrared imagery) and exploring cross-view learning could improve model robustness under different UAV flight attitudes. Lastly, extending GCRH to video-based detection with temporal encoders or memory modules can exploit cross-frame consistency to improve stability and

reduce false positives. Additionally, future work may consider rotation-equivariant modeling for arbitrarily oriented objects, self-supervised pretraining on large-scale aerial data for domain generalization, and standardized benchmarking of both accuracy and efficiency on real UAV platforms. By pursuing these directions, GCRH can evolve from a research prototype into a practical and broadly applicable system for real-time UAV vision applications.

CRediT authorship contribution statement

Hongcheng Xue: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation. **Tong Gao:** Writing – review & editing, Validation, Formal analysis. **Zhan Tang:** Validation, Supervision, Formal analysis. **Yuantian Xia:** Writing – review & editing, Validation, Supervision. **Longhe Wang:** Writing – review & editing, Visualization. **Lin Li:** Writing – review & editing, Validation, Supervision, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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